UNIVERSITY OF RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY

SCHOOL OF ICT

DEPARTMENT: COMPUTER ENGINEERING

MODULE: MOBILE APPLICATION SYSTEMS AND DESIGN

LAB 6: FLUTTER + FIREBASE CLOUD MESSAGING

DATE 19th MARCH 2026

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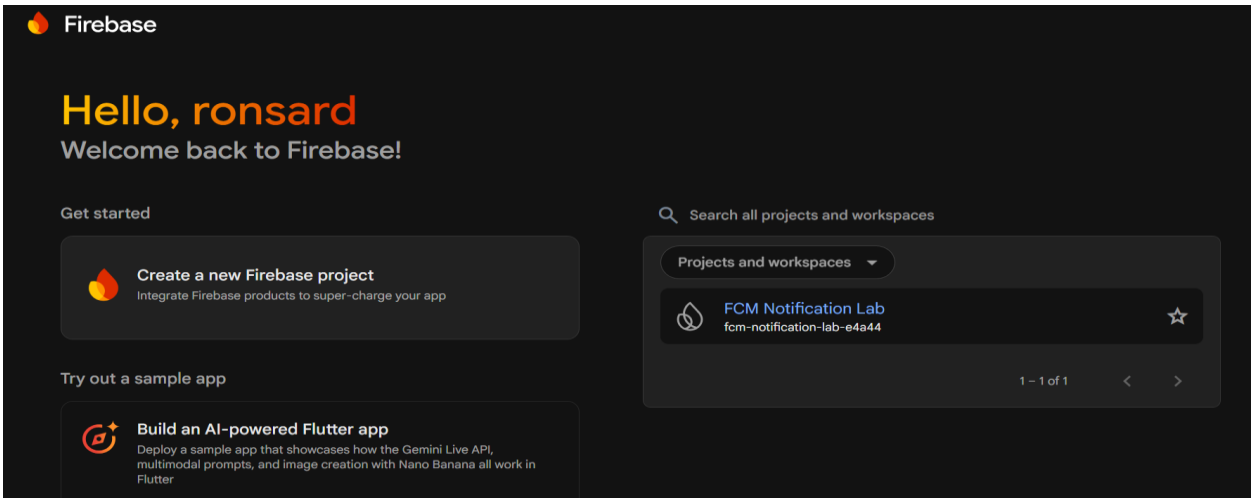
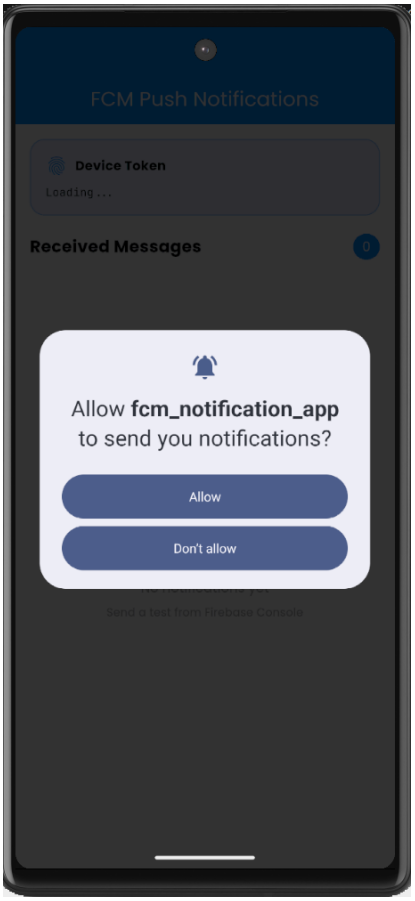
Section 1: Application Screenshots

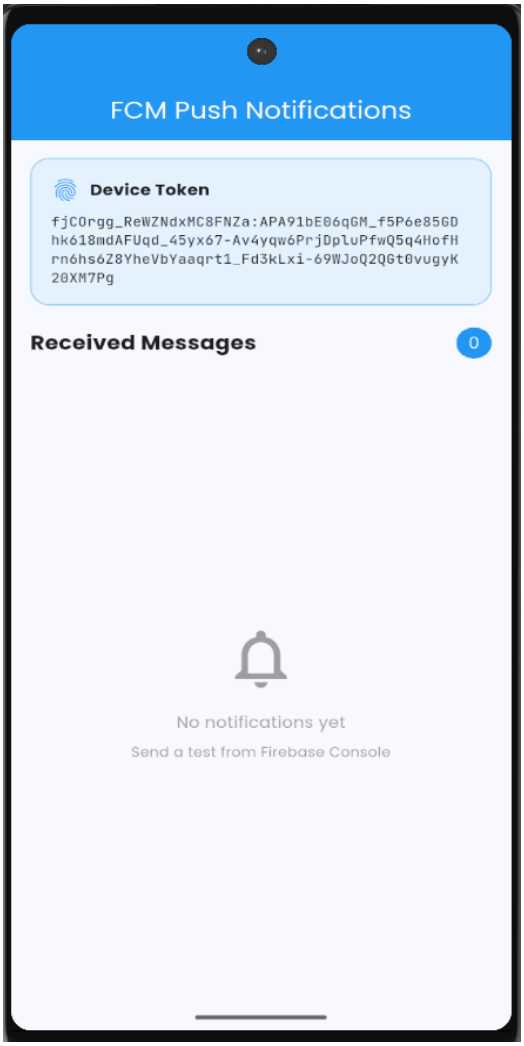
Firebase home page



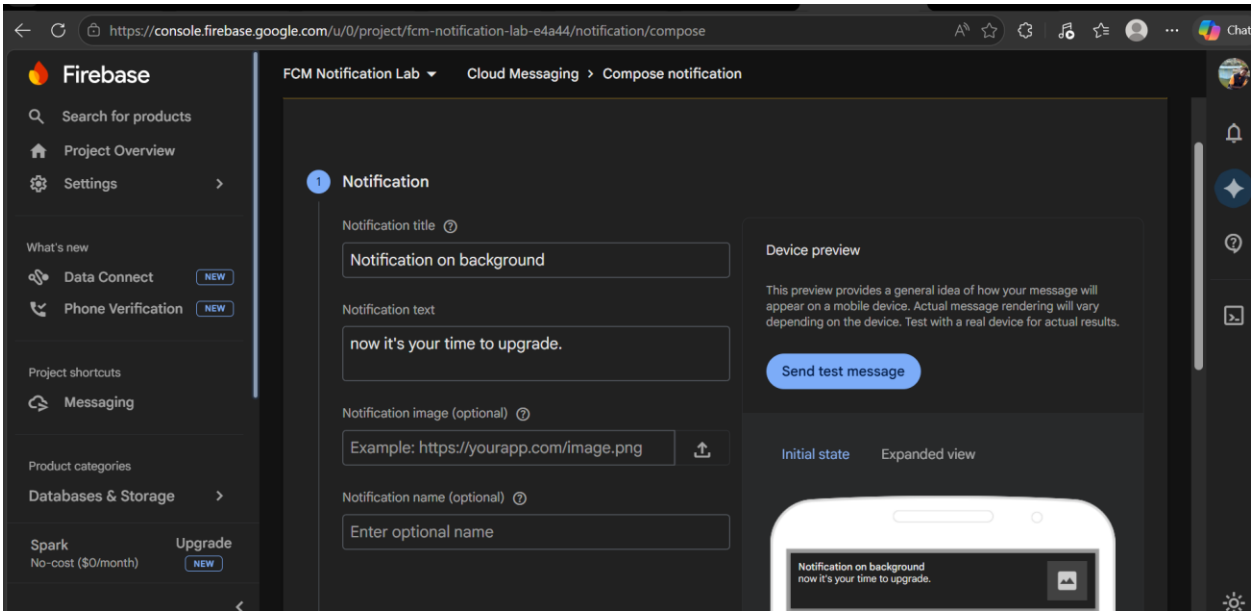
Notifications Enabling on Emulator



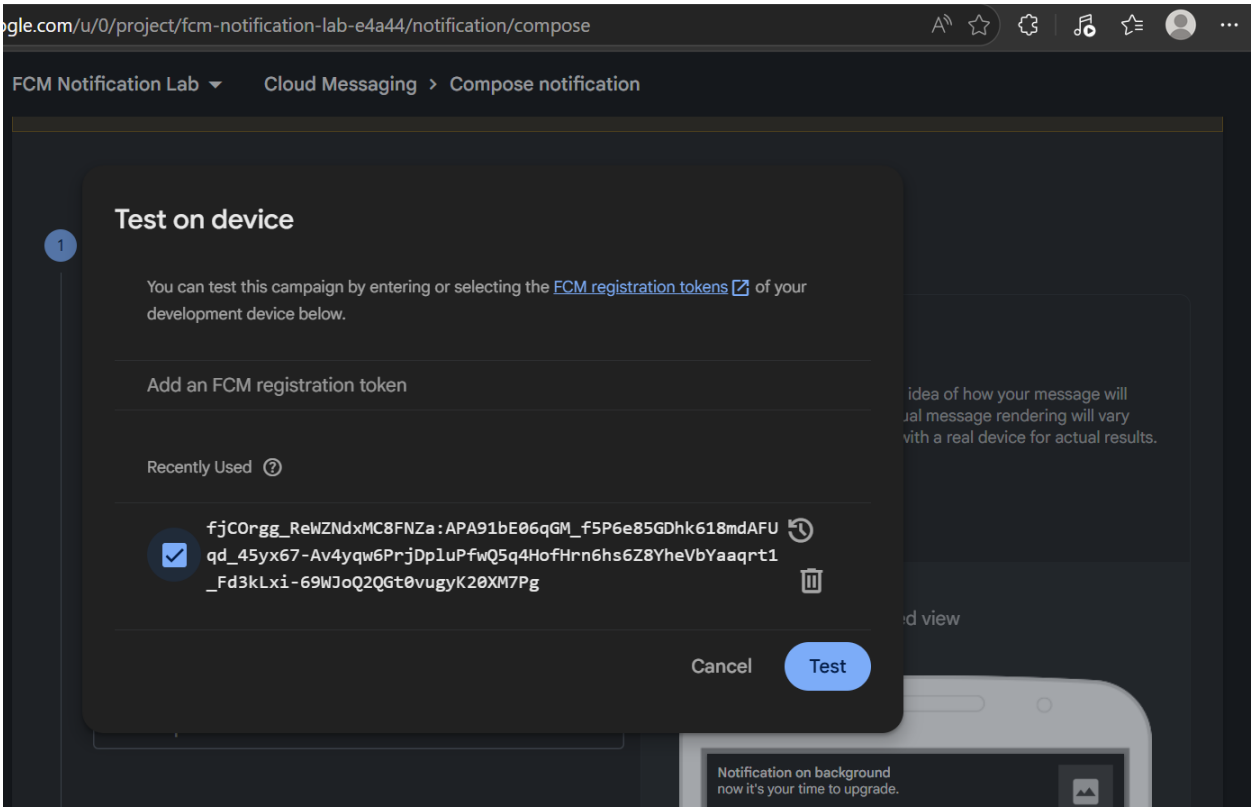
Home page for Flutter UI with token



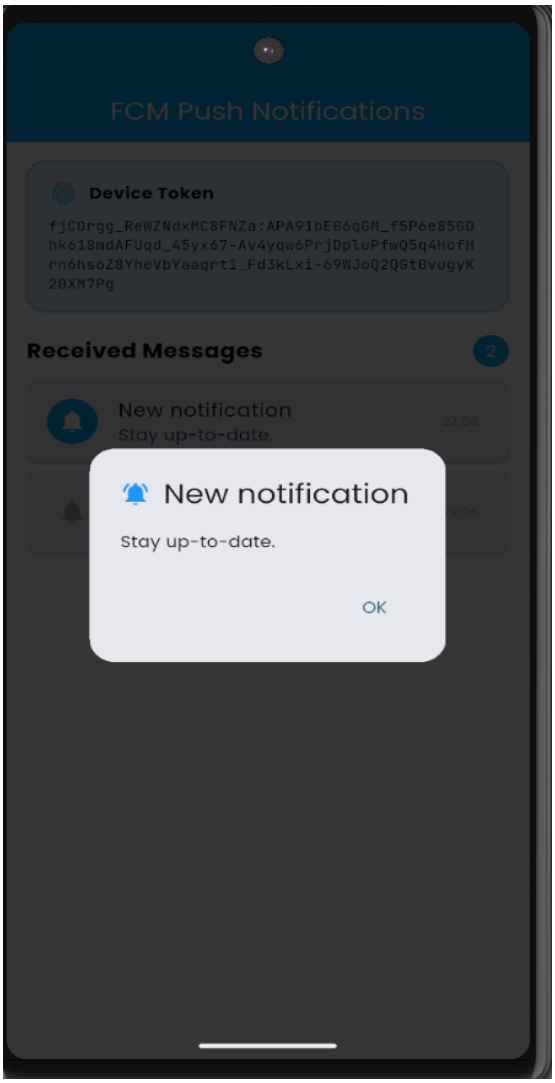
Cloud messaging interface



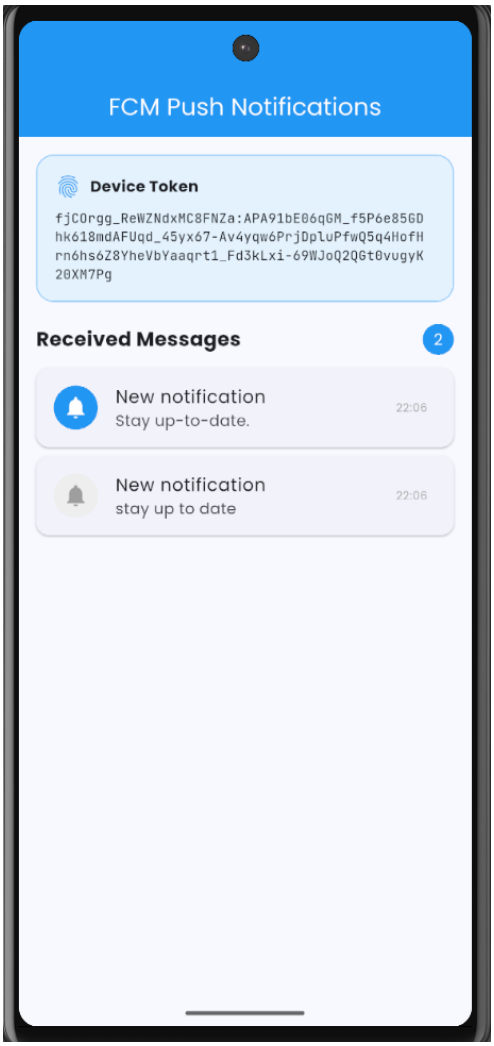
Test on Device using Given Token



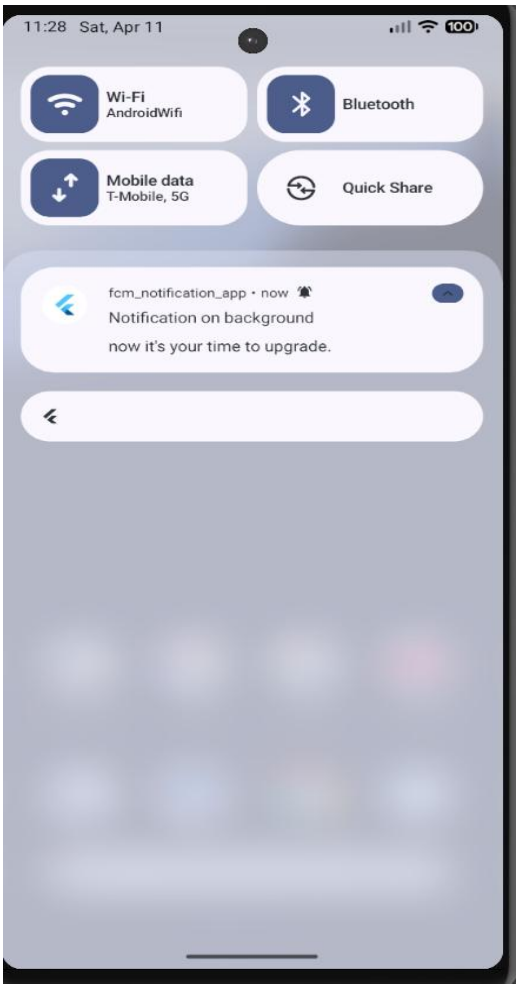
Notification Inside Application Popped up



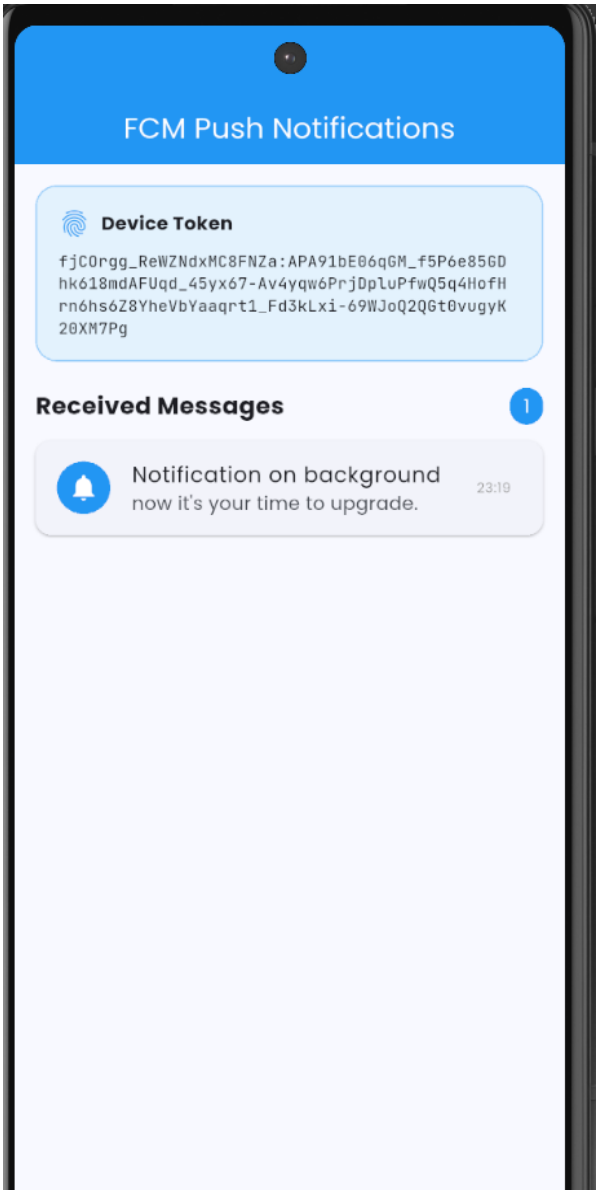
Screen with received Notification as Message



Notification Popped up on screen on background and in notification Place



Notification opened in Application shown in received message



Section 2: Observations & Key Steps (Handwritten)

UR - CST - GAKO
Group # 3
Handwritten Document

Observations

① How FCM works

- Firebase Cloud Messeging (FCM) is a cross - platform Messeging Solution
- each device / app installation gets a unique FCM registration token
- Messages are sent from firebase console to specific tokens
- FCM handles Messages delivery even when App is closed

2. Foreground vs Background Behaviour

App state	Behaviour	implementation required
Foreground	No auto system Notification, Must show custom popup	firebasemessaging.onmessage.listen()
Background	Auto shows in system tray; No popup	onbackgroundMessage handler
terminated	Same as background	getInitialMessage() to handle

③ Key challenges faced & Solution

Challenges	Solution
① Gradle build errors	upgrade AGP to 8.1.1 and gradle to 8.3
② Google service not found	added apply plugin: 'com.google.gms.google.services'
③ Background Messeges not working	added @pragma ('vm:entry-point') annotation

4. Important learnings

- FCM requires google play services on the device / emulator
- Android 13+ needs explicit notification permission request
- foreground messages need manual popup implementation
- the device token is unique and changes if app is cleared

Key Steps followed

Step 1 : · firebase project setup
- create firebase project "FCM notification lab"
- Regist android app with package name
- Download google service .json and placed in android/app

Step 2 : flutter dependencies
- added to pubspec.yaml

Step 3 : Gradle configuration
- android / build.gradle.kts - Added :
kotlin
classpath("com.google.gms:google-services:4.4.2")

Step 4 : Android permissions
→ added to AndroidManifest.XML :
<uses-permission android:name="android.permission.POST_NOTIFICATIONS"/>

Step 5 : Code implementation
→ As it has attached in the same directory on Github

Step : testing project
→ As it is shown in screenshot in document

CONCLUSION

Successfully implemented FCM push notifications in Flutter. The app can:

- Request and receive notification permissions
- Display unique device token
- Receive messages from Firebase Console
- Show popup notifications when app is open
- Display system tray notifications when app is in background
- Store and display received messages in app UI